# Safe System of Work

## Changing HGV and Trailer Wheels and Tyres

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## Overview

The following information provides guidance on the company's Safe System of Work for changing wheels and tyres on HGVs, tractor units and trailers, both in Fox Group depot workshops and off-site at customer sites or the roadside.

Vehicles slipping off jacks, collapsing supports and wheels falling from the hub can cause crushing injuries, entrapment and death. Wheel changes carried out off-site on unsuitable ground carry the greatest risk.

All employees have a responsibility for their safety and the safety of others by adhering to these safety systems.

The Operative must always refer to any associated Risk Assessment and Method Statements prior to starting works, including WF019 Working Under Vehicles, WF024 Tyre Maintenance and WF026 General Jacking and Blocking in the Fox Group Wagon Fitters Handbook (FG1.0).

## Hazards Identified

- Vehicle slipping off a jack or support
- Collapse of jacks, axle stands or packing
- Crushing and entrapment under the vehicle
- Wheel falling off the hub during lifting or removal
- Soft, uneven or sloping ground
- Vehicle rollaway or unexpected movement
- Moving traffic and plant at customer sites and the roadside
- Tyre explosion or rim failure during inflation
- High-pressure air lines
- Manual handling of heavy wheels
- Noise and vibration from impact tools
- Cuts, abrasions and trapped fingers
- Property damage to wheel arches, mudguards and lamps

## Personal Protective Equipment (PPE) Requirements










Gloves, safety footwear, hi-vis clothing and safety glasses must be worn at all times. Wear head protection where there is a risk of head injury, and hearing protection when using impact wrenches or compressors.

## Rules Before Lifting Any Vehicle

- Only trained, competent and authorised fitters may jack vehicles or change wheels;
- Complete a point of work risk assessment before starting. At customer sites, agree the work location with the customer and follow their site rules;
- Only lift on firm, level hardstanding. If the ground is soft, uneven or sloping, STOP and arrange for the vehicle to be moved to a safe location before any lift;
- Apply the parking brake, switch the engine off and remove the keys. Display “Do Not Drive / Work in Progress” at the cab;
- Segregate the work area with cones and barriers. Use hazard lights and a beacon at the roadside;
- Use only jacks, axle stands and base plates that are rated for the load, visually inspected before use and in date for thorough examination (LOLER);
- Jacks and stands must only be placed at the vehicle manufacturer’s jacking points;
- Never use sockets, tools, bricks or other improvised items as jack spacers or packing;
- Never rely on a jack alone. Fit axle stands before removing a wheel, and never go under a vehicle supported only by jacks;
- Where safe jacking cannot be achieved on site, use the approved tyre contractor.

## How to Change a Wheel Correctly

**Step 1 – Choose a safe location and segregate the area.**
Position the vehicle on firm, level hardstanding away from traffic. Set out cones and barriers around the

work area. Parking brake on, engine off, keys removed.



Illustrative image

**Step 2 – Chock the wheels.**
Chock the wheels that will stay on the ground, on the opposite side and axle to the wheel being changed, to stop any movement.



Illustrative image

**Step 3 – Loosen the wheel nuts with the wheel on the ground.**
Crack each wheel nut loose by about a half turn while the tyre is still carrying load on the ground. Do

NOT remove the nuts at this stage – they must keep the wheel on the hub while the vehicle is lifted.



Illustrative image

**Step 4 – Jack at the manufacturer's jacking point.**

Place a rated jack on a flat steel base plate, directly under the manufacturer's jacking point (normally the axle beam). Operate the jack from the side, never from underneath the vehicle.



Illustrative image

**Step 5 – Raise the wheel and fit an axle stand.**

Raise the axle until the tyre is just clear of the ground. Fit a rated axle stand with the correct locking pin

under the axle next to the jack. Lower the load onto the stand. Do not rely on the jack alone.



Illustrative image

**Step 6 – Remove the nuts and the wheel.**
Only now remove the wheel nuts. Use a wheel dolly or lifter to support the wheel and roll it off the studs. Stay to the side and never put any part of your body under the vehicle.



Illustrative image

**Step 7 – Refit the wheel and torque the nuts.**
Clean the hub and studs and check them for damage. Fit the replacement wheel and run the nuts on by hand. Lower the vehicle, remove the stand and jack, then tighten the nuts in a diagonal (star) pattern to

the manufacturer's specified torque using a calibrated torque wrench. Fit wheel nut indicators.



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**Step 8 – Check the tyre pressure and re-check the torque.**
Inflate to the manufacturer's pressure using a clip-on chuck with a long hose and in-line gauge, standing outside the line of the tyre wall. Any tyre removed from its rim must be inflated in a safety cage. Re-check the wheel nut torque after the settling period stated by the manufacturer and record the work.



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## Do's and Don'ts When Changing Wheels

### Do's

- **Complete a point of work risk assessment.**
- **Lift only on firm, level hardstanding.**
- **Chock the wheels that stay on the ground.**
- **Loosen nuts with the wheel on the ground.**
- **Use rated, inspected and in-date jacks and stands.**
- **Use base plates under jacks and stands.**
- **Jack only at manufacturer's jacking points.**
- **Fit axle stands before removing the wheel.**
- **Use a wheel dolly or lifter for heavy wheels.**

- Torque nuts with a calibrated torque wrench.
- Re-check torque after the settling period.
- Stand outside the tyre wall line when inflating.

## Don’ts

- Never remove all wheel nuts before lifting.
- Never go under a vehicle supported only by jacks.
- Never use sockets, tools or bricks as jack spacers.
- Never jack on soft, uneven or sloping ground.
- Never jack on bumpers, underrun bars or body panels.
- Never exceed the safe working load of a jack or stand.
- Never inflate a tyre fitted to a rim without a cage.